

Appendix C: Automated Test Suite Inventory

This appendix lists the test categories behind the counts reported in `??`. Every test uses real mathematical computations—no mocks or fakes.

- ▶ **Categorical axiom tests:** Identity morphism existence, composition associativity, weight invariants, `is_well_formed()` full axiom check
- ▶ **Enriched category tests:** Hom-value constraints, composition inequality, categorical magnitude, magnitude deficit, full composition check, role clustering
- ▶ **Diagram type tests:** Pregroup diagrams validated for `dom == Ty()` and `cod == s`, correct box counts, diagram equality
- ▶ **Metrics tests:** Normal form preservation, depth computation with graceful fallback for pregroup diagrams
- ▶ **Natural transformation tests** (20 tests): Component morphism construction, `naturality_holds / verify_naturality` on identity and incomplete maps, identity transformation generation, vertical composition of transformations, completeness checking over all domain